

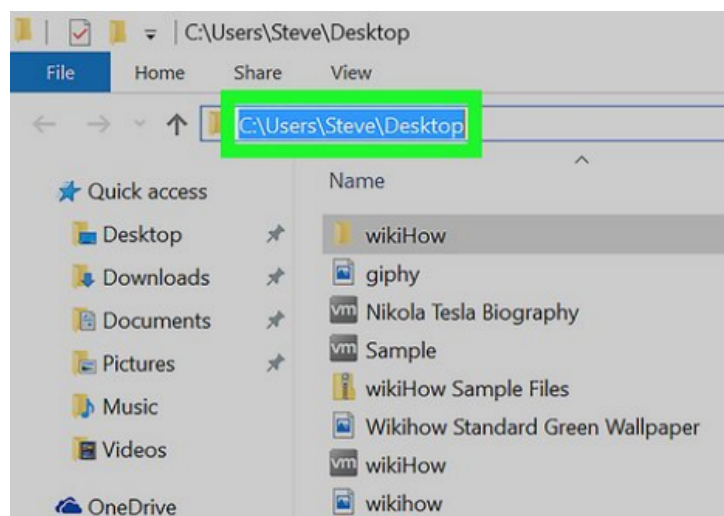
26F Stat TA Session W2

0. Preparation for TA session

- Please download the W2 example code and the attached datasets [click here](#)
 - the datasets can also be found in [OpenIntro](#)

1. Path/ Directory

- Path: the place where a file is saved



- Working directory: the place where `R` reads in your files
- Use `getwd()` to find your current working directory
- Use `setwd()` to change your working directory



Note when you change the directory, remember to replace `\` with `/`

```
setwd("C:\your\working\directory") #Error!  
setwd("C:\\your\\working\\directory") #Works on Windows!  
setwd("C:/your/working/directory") #Works everywhere!
```

▼ Q. Why we can't simply copy the path and use `\` ?

Ans. Because `\` is an escape character in `R`

An escape character is used to represent special characters within a string

e.g.,

- `\n`: newline
- `\t`: tab

If you want to type `\` in `R`, use `\\`

2. Read Files

- The first step to perform statistical analysis is to read in files!
- There are several functions in `R` for reading different data formats

```
df <- read.csv("file.csv") #Work if file in working directory
df <- read.csv("folder_name/file.csv") #If file is in folder
df <- read.csv("C:/your/working/directory/file.csv")

# example
county <- read.csv("county.csv")
loan <- read.csv("loan50.csv")
```

3. What's in the dataset?

What the variables represent?

- `county.csv`
 - `county`: county name in the US
 - `homeownership`: home ownership rate during 2006-2010
 - `multi_unit`: percent of housing units in multi-unit structures, 2006-2010
- `loan50.csv`
 - `annual_income`: annual income of the applicant
 - `interest_rate`: interest rate of the loan the applicant received
 - `grade`: grade associated with the loan
- You can also find detailed descriptions in [OpenIntro](#)

What's hidden behind the numbers?

- You may want to first look at the numbers
 - What the average?
 - How dispersed the data is?
 - Is there any outliers? (離群值)

- Is there any recording error in the dataset?
- **Which method should we apply for further analysis?**
- One good way to get familiar with your dataset is to **draw figures** for variables of your interests
- To do this, we can use the package `ggplot2`

Scatter Plot `geom_point()`

- Check if two variables are related

```
ggplot(county, aes(x = multi_unit, y = homeownership)) +  
  geom_point(color = "steelblue",  
             size = 1,           # the arguments are to beautify your p  
lot  
             shape = 5  
            )
```

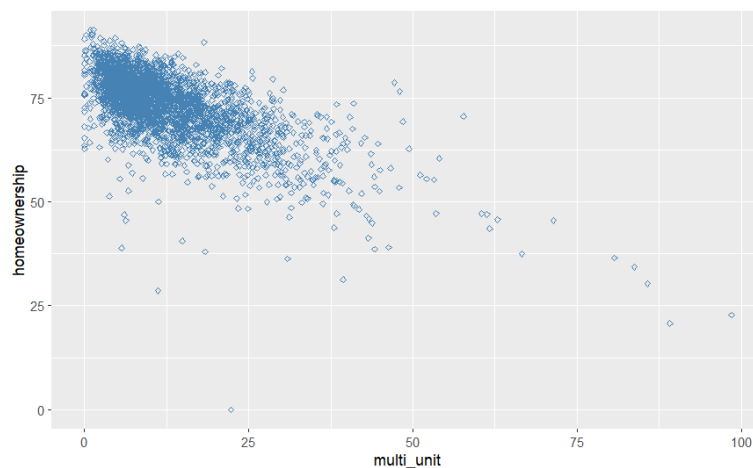


Figure: Scatter plot of multiple unit ratio and home ownerships

Histogram `geom_histogram()`

- How a variable is distributed? Is it skewed?

```
ggplot(loan, aes(x = interest_rate)) +  
  geom_histogram(binwidth = 2.5, ## choose appropriate bin width  
                fill = "steelblue",  
                color = "black"  
               )
```

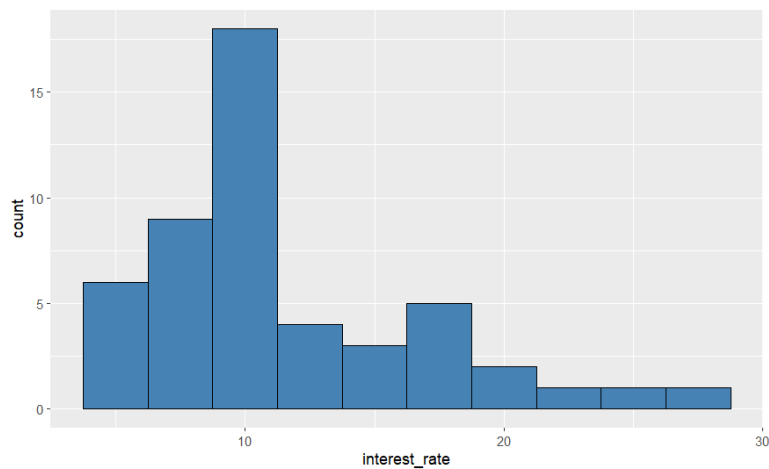


Figure: Histogram of interest rates

Box plot `geom_boxplot()`

- How dispersed a variable is? Is there any outliers?

```
ggplot(loan, aes(x = grade, y = interest_rate)) +
  geom_boxplot(width = 0.3)
```

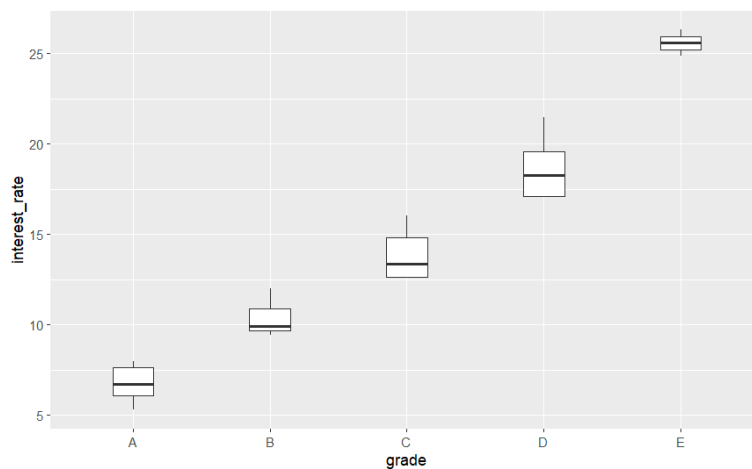


Figure: Box plot of interest rates by verified income

Dot plot `geom_dotplot()`

- Observe all data points, especially in small datasets

```
ggplot(loan, aes(x = grade, y = interest_rate)) +
  geom_dotplot(binaxis = 'y',
    stackdir = 'center',
    dotsize = 0.5,
    fill = "steelblue"
  )
```

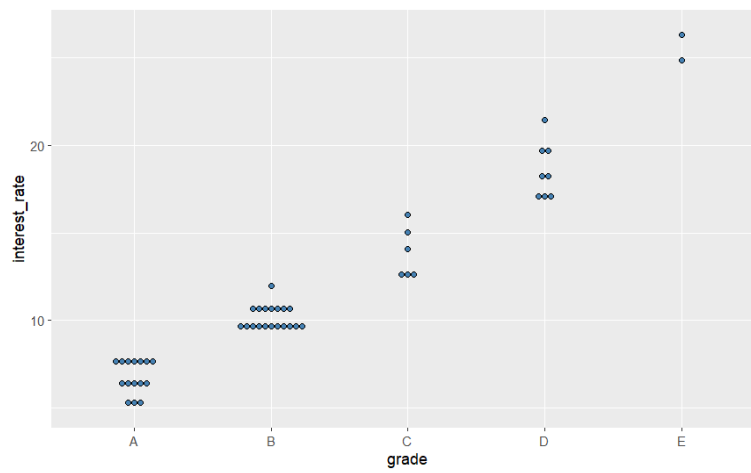


Figure: Dot plot of interest rates by verified income

Exercises

W2 Exercises